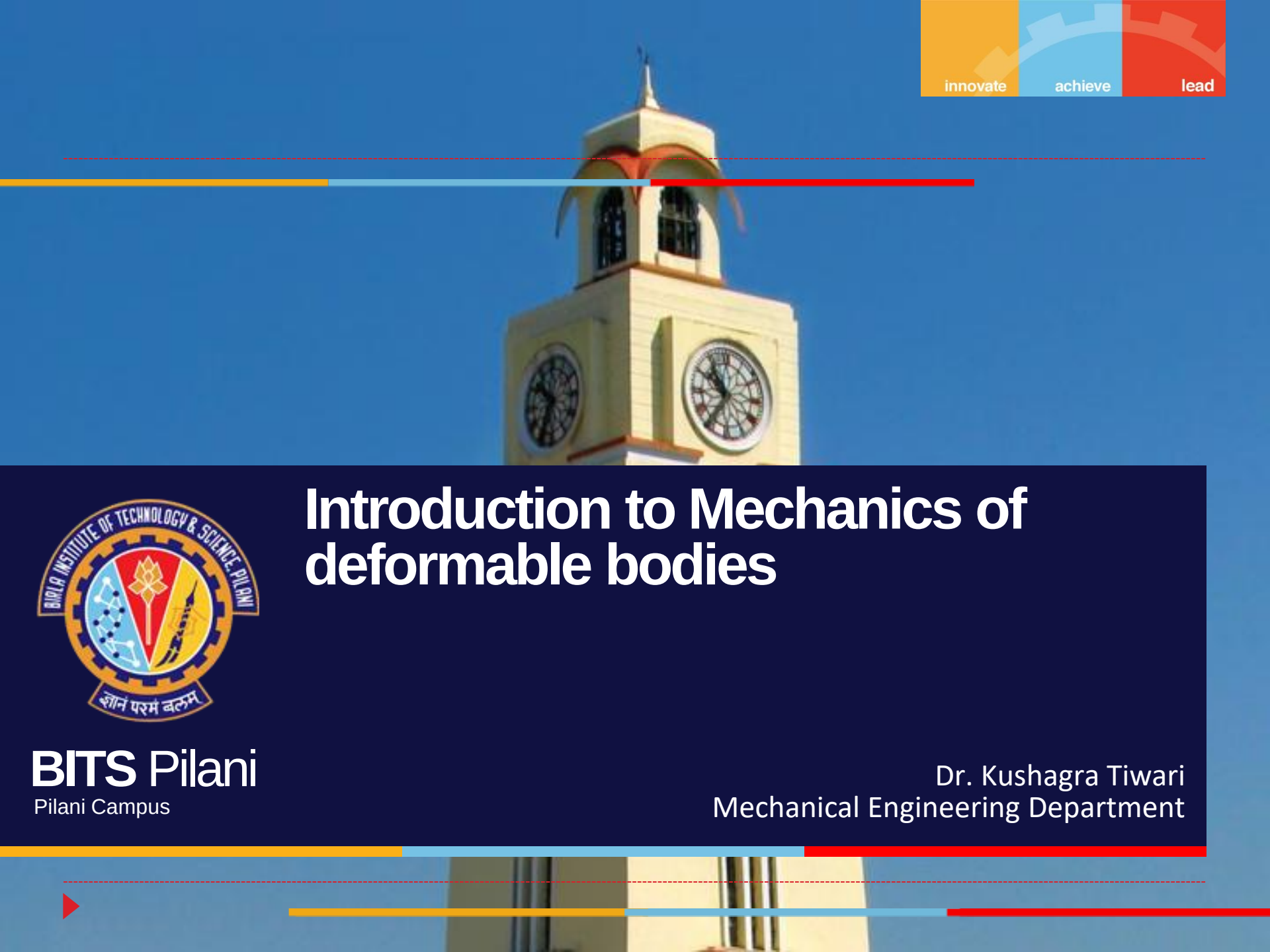


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Introduction to Mechanics of deformable bodies

BITS Pilani
Pilani Campus

Dr. Kushagra Tiwari
Mechanical Engineering Department

CHAPTER 3

FORCE AND MOMENTS TRANSMITTED BY SLENDER MEMBERS

Forces and Moments Transmitted by Slender Members

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Slender members

These are load carrying elements having much greater length (at least five times) than its lateral dimensions. Slender members can be pulled, bent and twisted

Examples: beam, columns, shafts, rods, stringers, struts, and links.

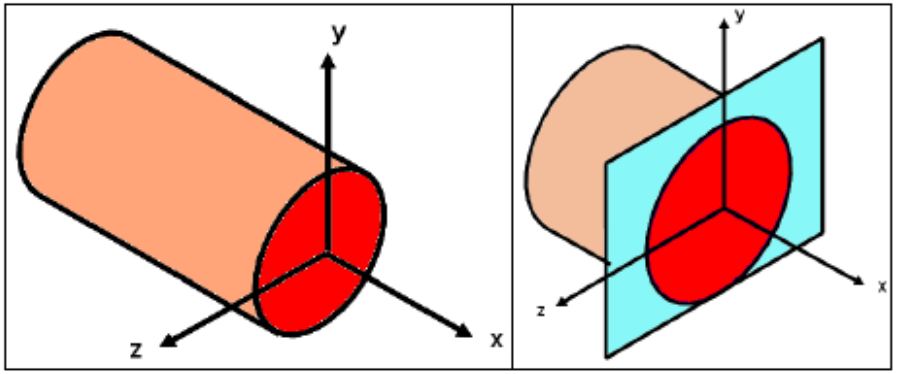




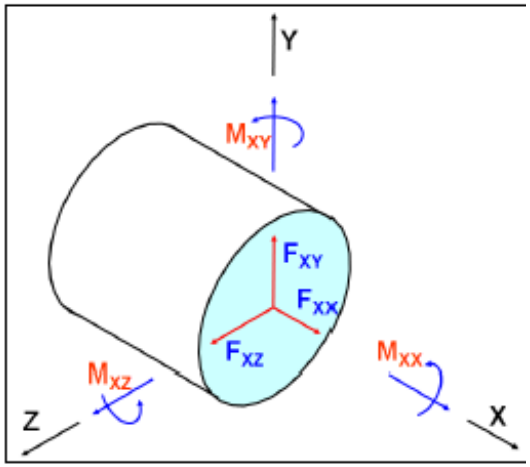
GENERAL METHOD TO FIND INTERNAL FORCES AND MOMENTS

- A general method for determining the internal forces and moments acting across any section of a slender member which is in equilibrium is to cut and that part & isolate from the system.
- The isolated part is in equilibrium. Apply the equations of equilibrium to find internal forces and moments.

- x- Section : The section normal to x- axis
- y- section : The section normal to y- axis
- z- section : The section normal to z- axis



Forces and moments acting on a cross section of a member



- F_{XX}: Force acting on x-section and is along x- axis
- Similarly F_{XY}, F_{XZ}, M_{XX}, M_{XY} and M_{XZ} can be explained

NOMENCLATURE

- F_{XX} (Axial force) : This component tends to elongate the member and is often given the symbol *F* or F_x.
- F_{XY} & F_{XZ} (Shear force) : These components tend to shear one part of the member relative to the adjacent part and are often given the symbols V, or V_Y and V_Z

- M_{XX} (Twisting moment) : This component is responsible for the twisting of the member about its axis and is often given the symbol M_T or M_{TX}.
- M_{XY} & M_{XZ} (Bending moments): These components cause the member to bend and are often given the symbols M_b, or M_{by} and M_{bz}.

Positive face of given section

If the outward normal points in a positive coordinate direction then that face is called as positive face

Negative face of given section

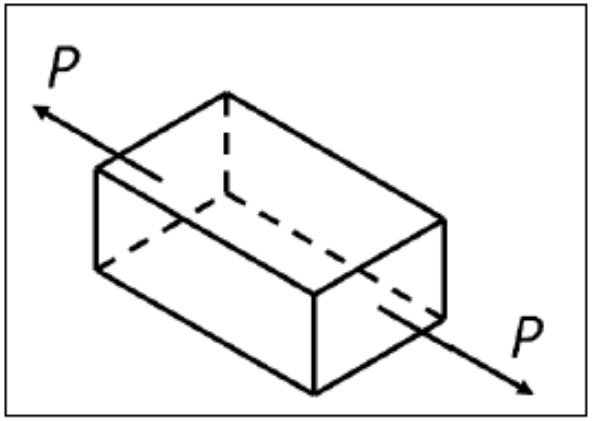
If the outward normal points in a negative coordinate direction then that face is called as Negative face



Sign convention for the axial force, shear force, and bending moment.

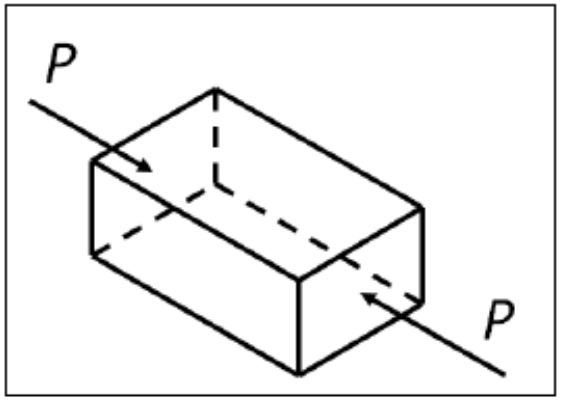
- If force or moment component acts on a positive face in a positive coordinate direction then these components are treated as *positive*
- If force or moment component acts on a negative face in a negative coordinate direction then these components are treated as *positive*

Positive Force (Tensile)



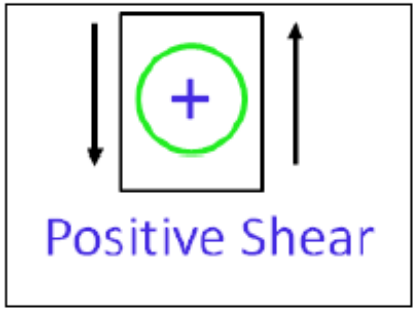
- If force or moment component acts on a positive face in a negative coordinate direction then these components are treated as *negative*
- If force or moment component acts on a negative face in a positive coordinate direction then these components are treated as *negative*

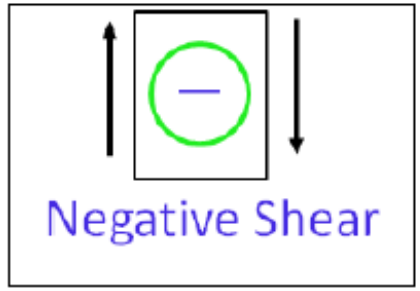
Negative Force (Compressive)



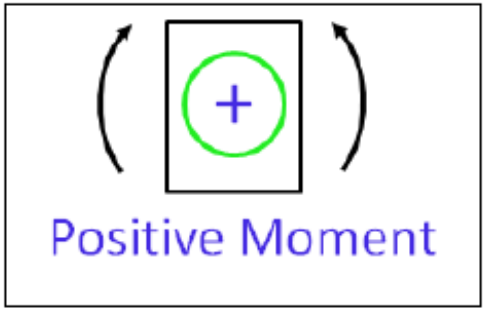
Sign convention for the axial force, shear force, and bending moment.

Shear force (V)

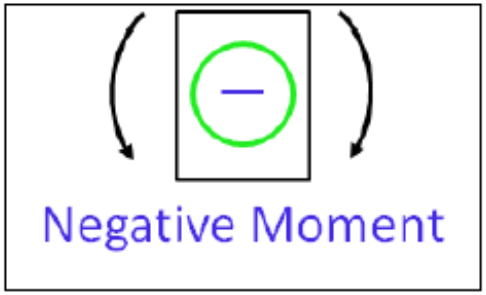




Bending moment (M_b)



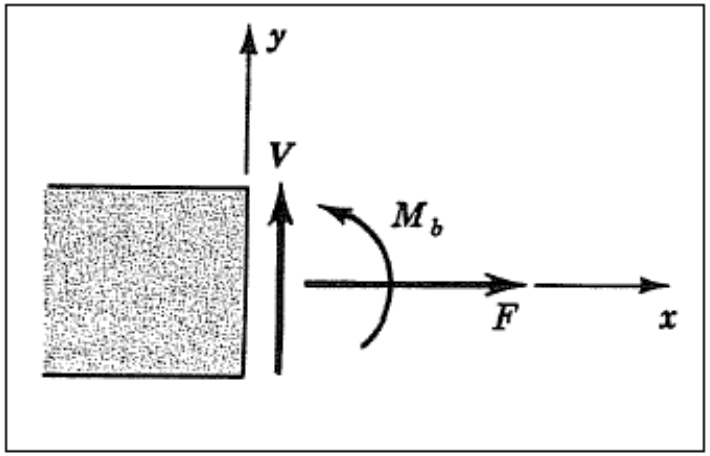
Sagging



Hogging

If the plane of loading is the *x*-*y* plane then only three components occur:

- The axial force F_{xx} (F), the shear force F_{xy}, (V), and the bending moment M_{xz} (M_b),



Force and moment components in two dimensions

The steps involved in solving the problems

1. Idealize the actual problem, i.e., create a model of the system, and isolate the main structure, showing all forces acting on the structure.
2. Using the equations of equilibrium calculate unknown external forces or support reactions

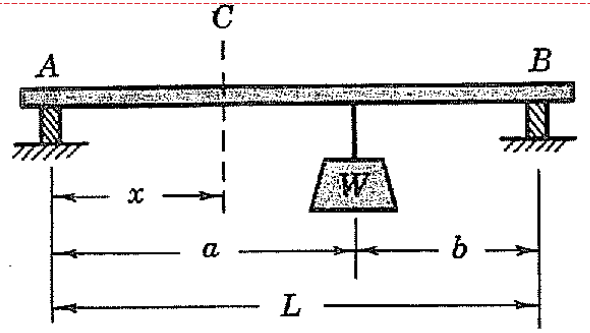
$$\Sigma F = 0 \quad \& \quad \Sigma M = 0$$

3. Cut the member at a point of interest, isolate one of the segments, and repeat step 2 on that segment.

Ex 3.1

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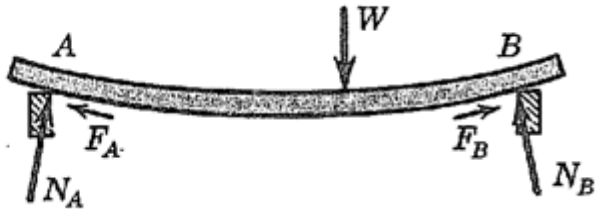
Find the forces and moments acting at section C



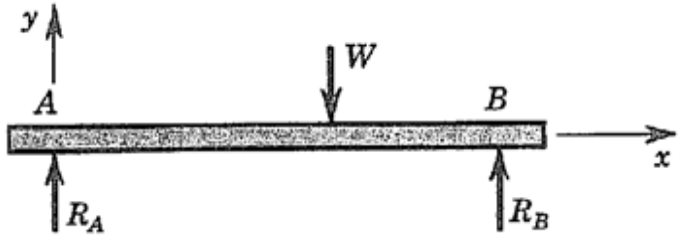
(a)

Assumptions

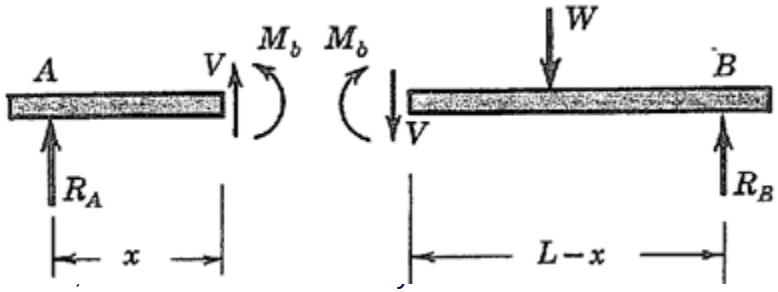
- F_A, F_B neglected
- Self weight of the beam neglected



(b)



(c)



(d)

Shear force diagram

Graph which shows shear force plotted against distance along a beam

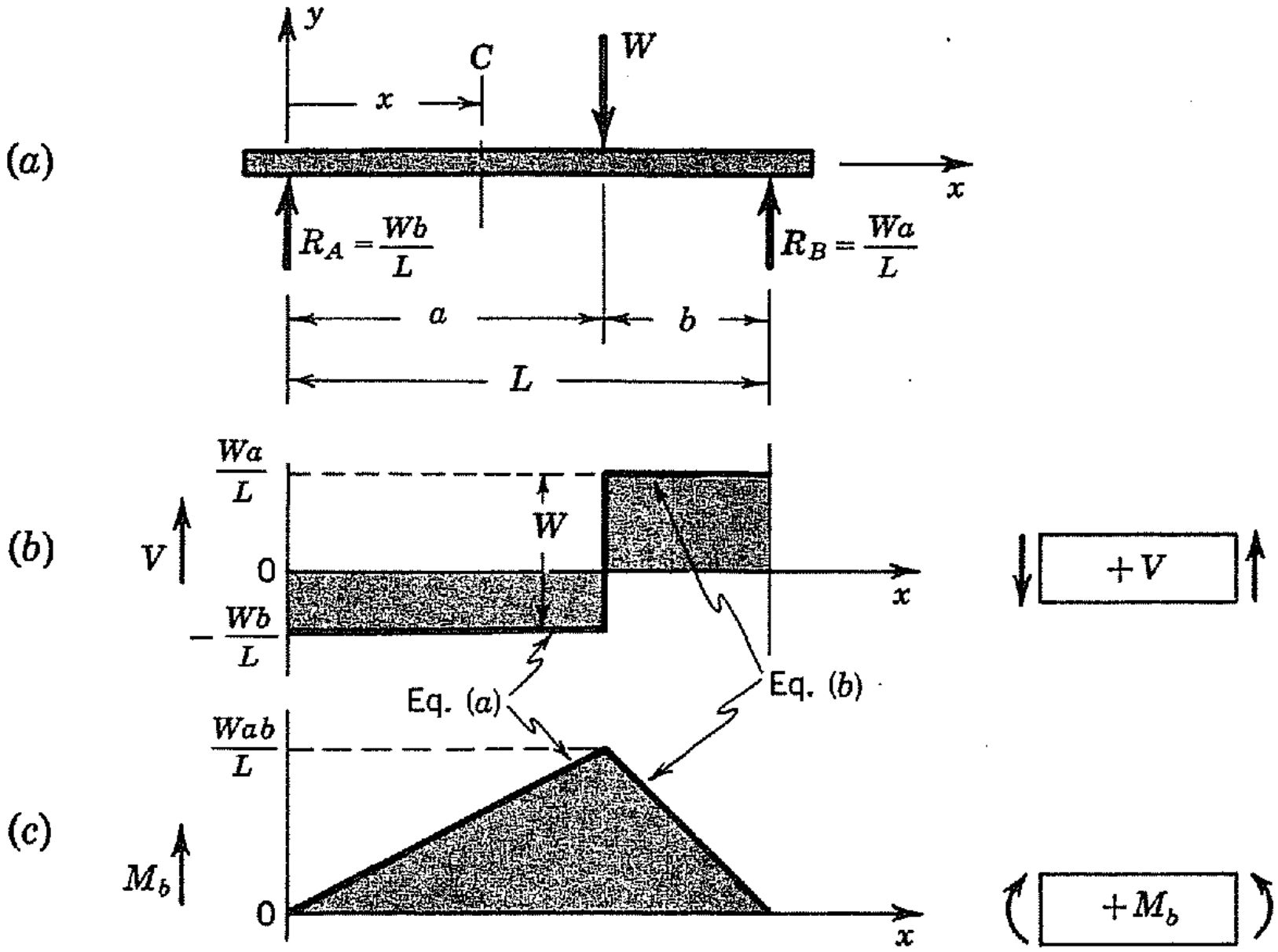
Bending moment diagram

Graph which shows bending moment plotted against distance along a beam

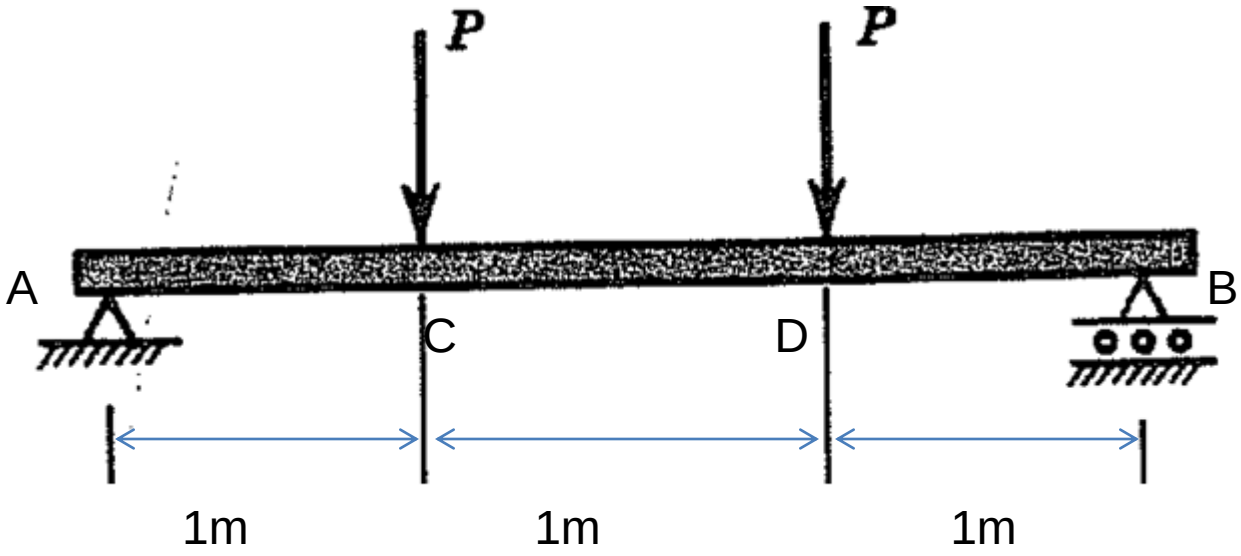
Point of Contra flexure [Inflection point]:

It is the point on the bending moment diagram where bending moment changes the sign from positive to negative or vice versa.

It is also called 'Inflection point'. At the point of inflection point or contra flexure the bending moment is zero.

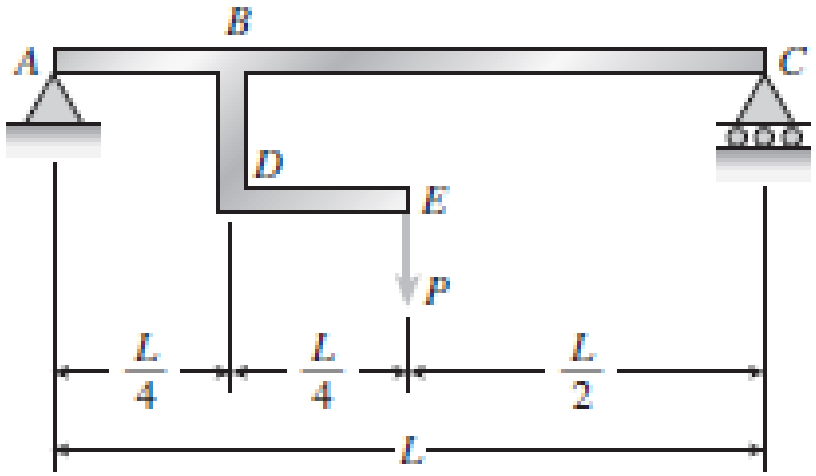

(a)
y
C
W
x
x
$R_A = \frac{Wb}{L}$
$R_B = \frac{Wa}{L}$
a
b
L
(b)
$\frac{Wa}{L}$
V
W
0
x
$-\frac{Wb}{L}$
Eq. (a)
Eq. (b)
+V
(c)
$\frac{Wab}{L}$
M_b
0
x
$+M_b$

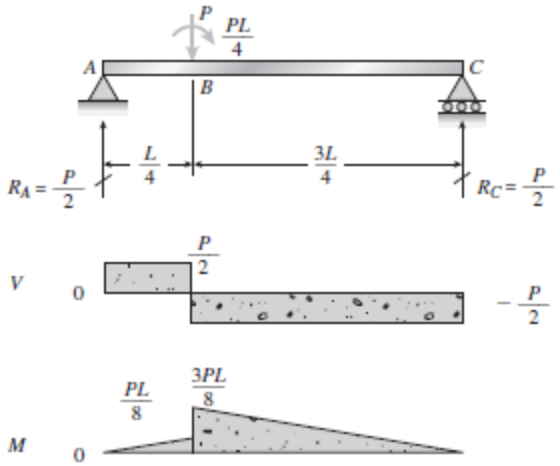
P= 10kN



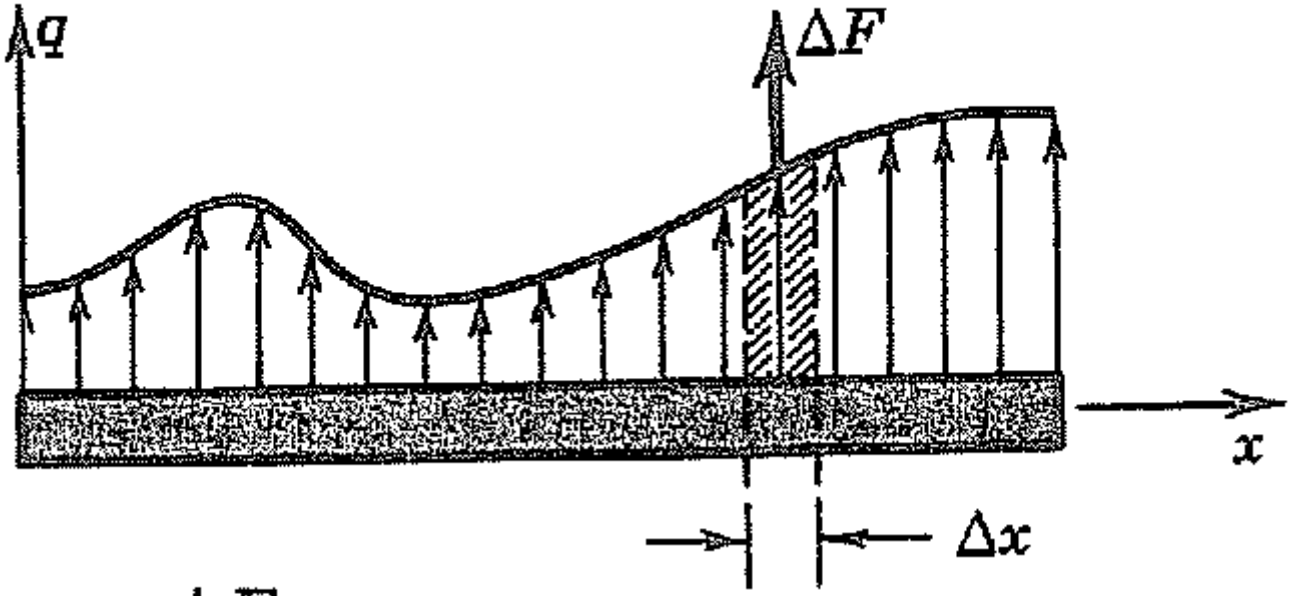
A simply supported beam ABC is loaded by a vertical load P acting at the end of a bracket BDE (see figure).

Draw the shear-force and bending-moment diagrams for beam ABC.





Distributed loading



$$q = \lim_{\Delta x \to 0} \frac{\Delta F}{\Delta x}$$

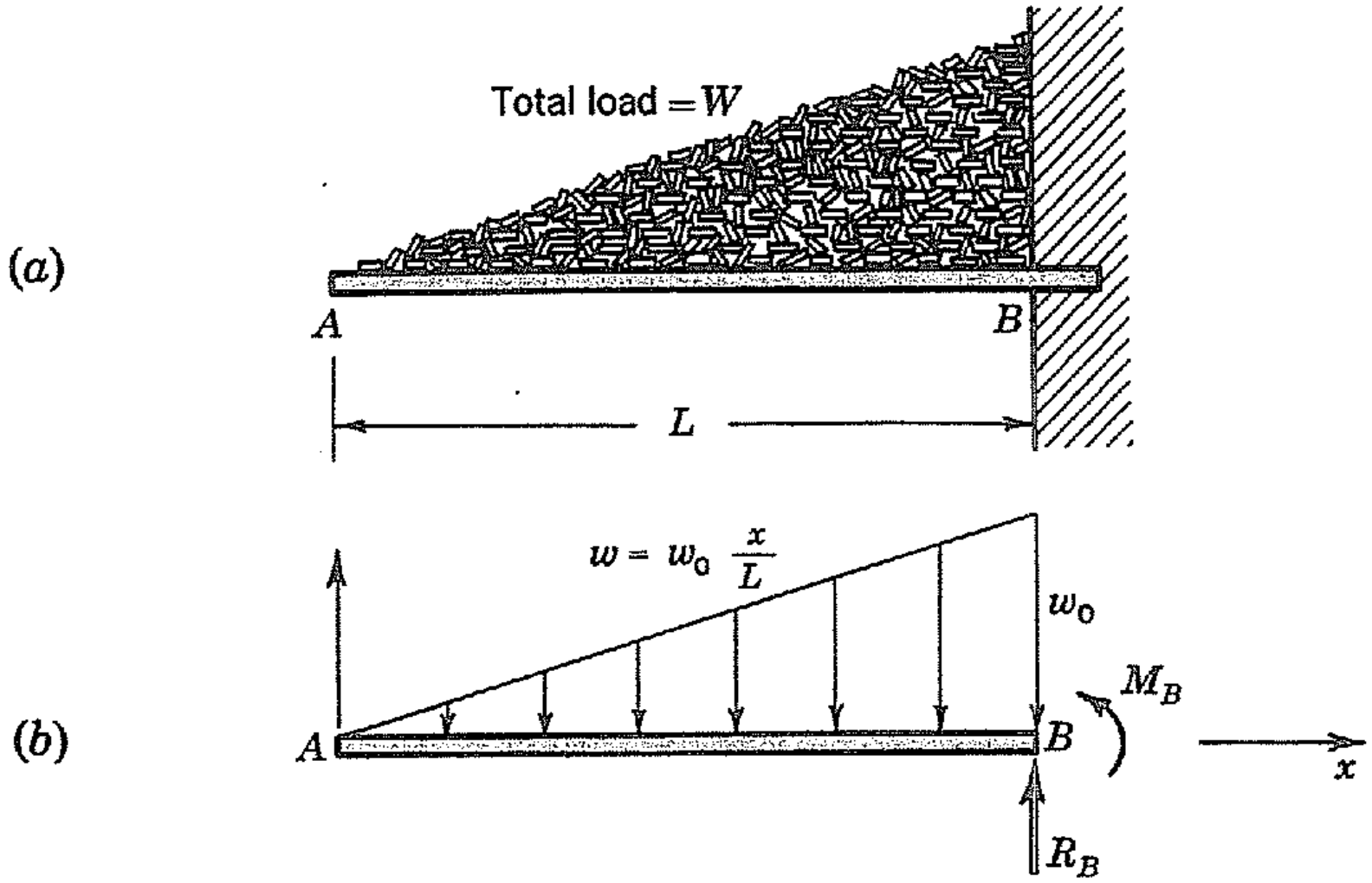
In engineering the most common distribution of loading are:

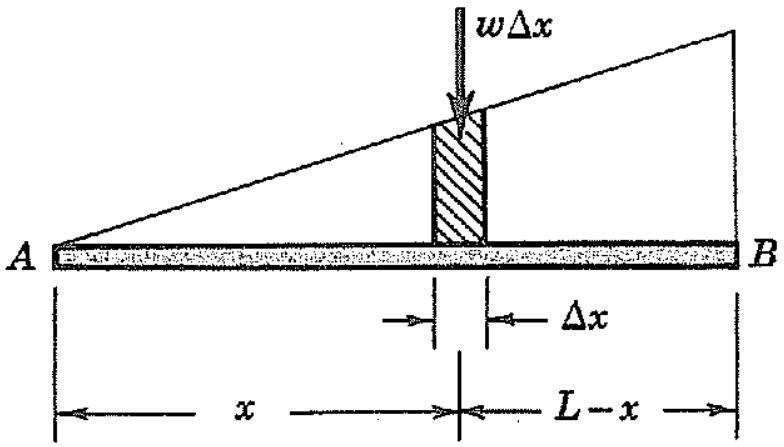
a) uniform distribution where q(x) is constant.
b) linearly varying distribution where q(x)= Ax+B

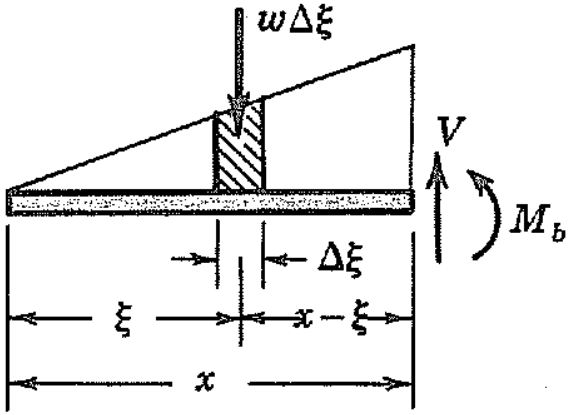
SIGN CONVENTION

Loading is positive when it acts in the direction of positive coordinate axis.

Distributed loads



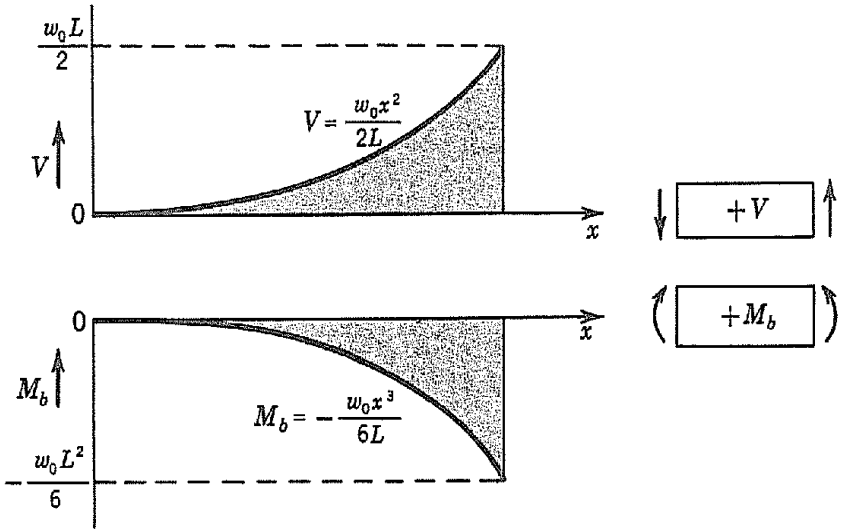




$$V = w_o \frac{x^2}{2L}$$

$$M_b = -w_o \frac{x^3}{6L}$$

which are valid for any value of x between 0 and L

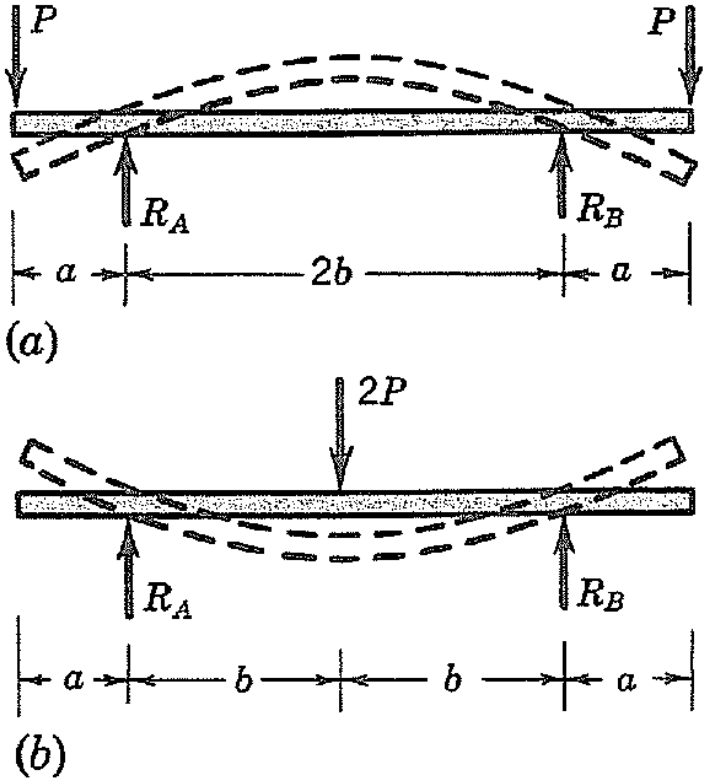


RESULTANT OF DISTRIBUTED LOADING

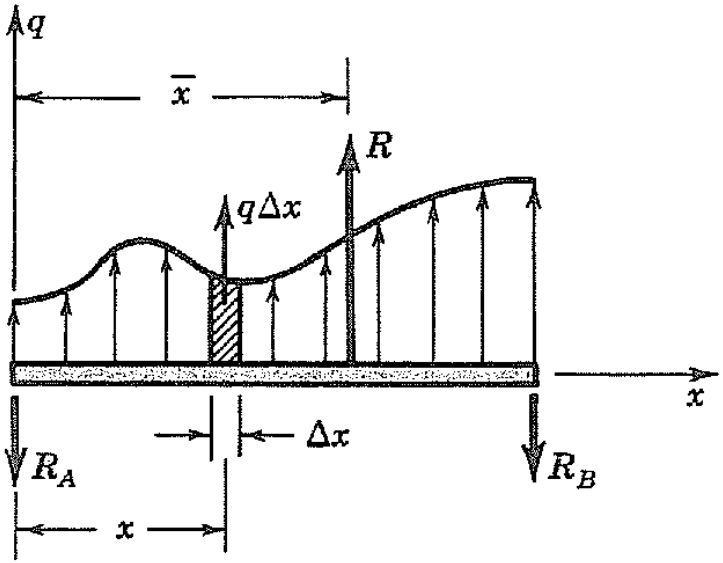
- More convenient to work with rather than with actual distribution.

Limitations:

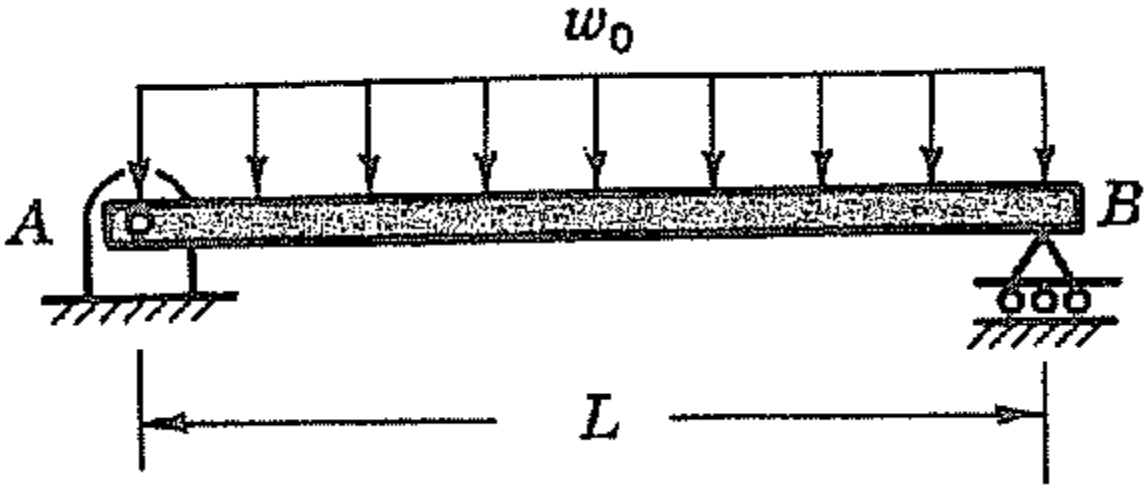
1. permissible when only evaluating external reactions on the members.
2. mot allowed to calculate internal forces and moments.



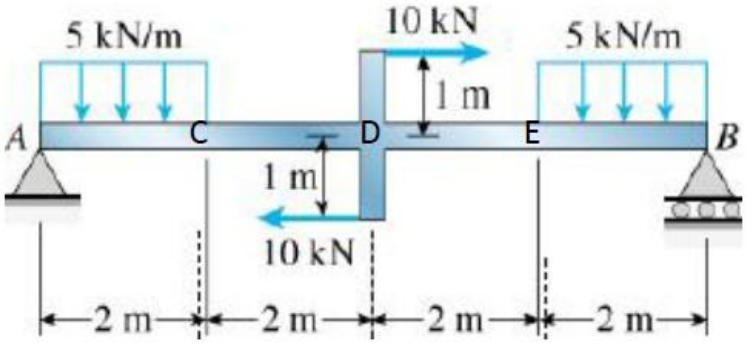
Resultant and its location for the given distributed loading



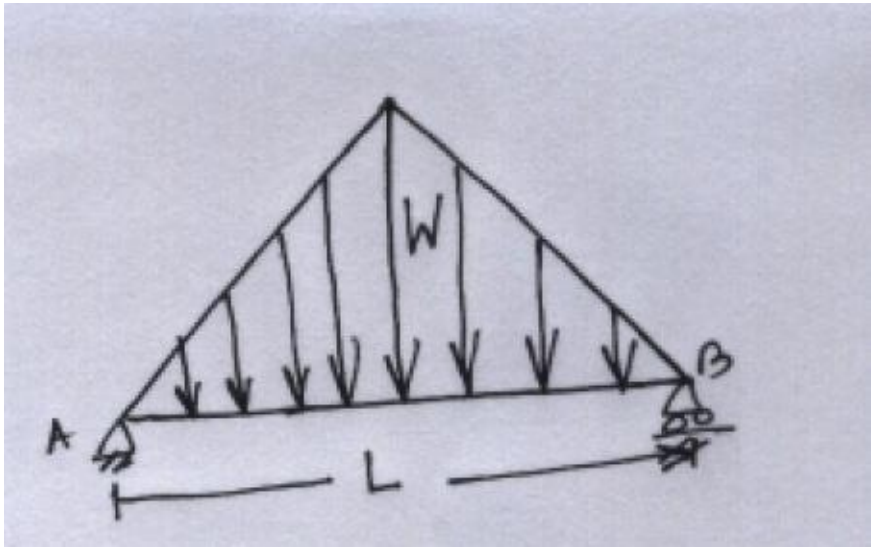
$$R = \int_0^L q\,dx \qquad \text{and} \qquad \bar{x} = \frac{\int_0^L xq\,dx}{R}$$

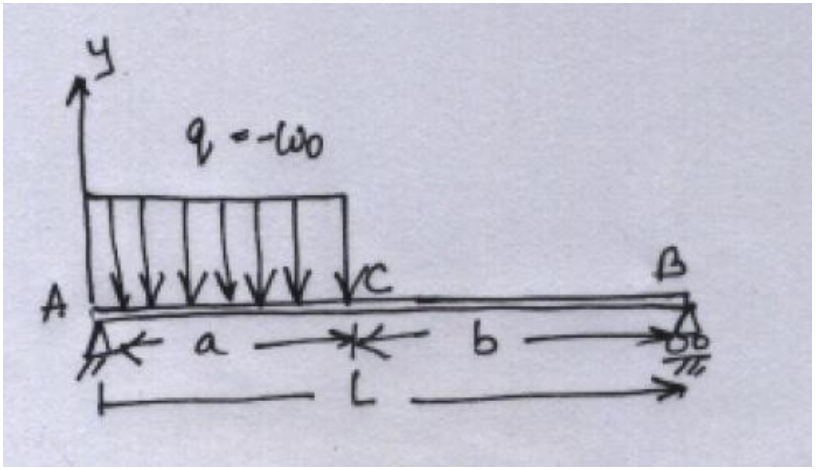

w_0
A
B
L

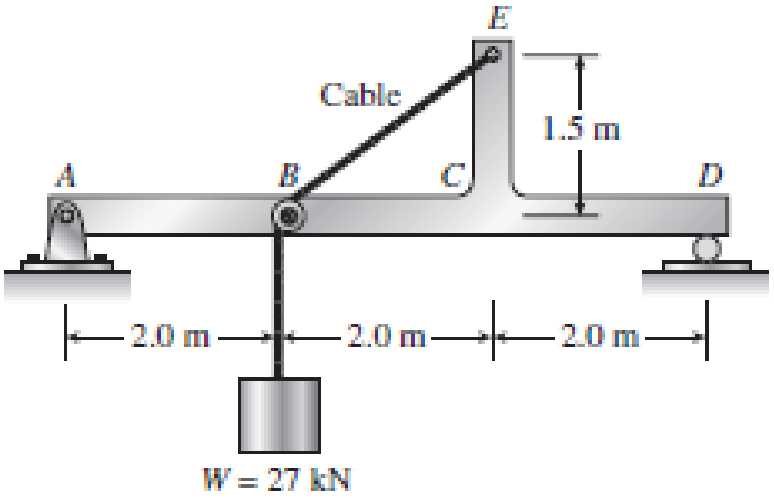
Draw the shear force and bending moment diagram (mention the sign convention) for the beam AB as shown in figure. *Write the values of shear force and bending moment at salient points clearly in a table.* Also write the values of shear force and bending moment at salient points clearly in the shear force and bending moment diagram. ***If any discontinuity point is left in the table please add it.***



	A	C	D		E		B
			D-	D+	E-	E+	
V (kN)							
M_b(kNm)							


W
A
B
L


y
q = -w0
A
C
B
a
b
L

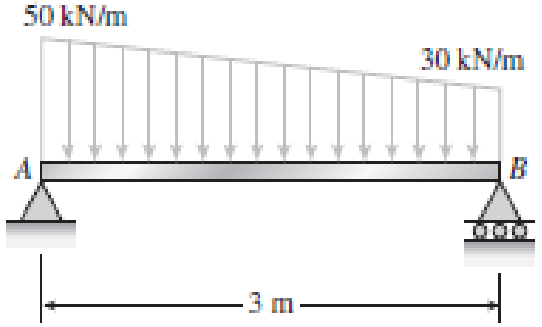


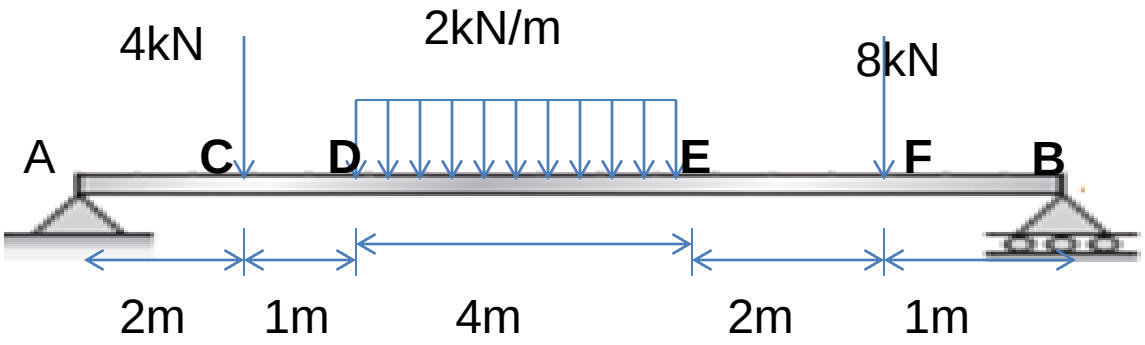
The simply-supported beam *ABCD* is loaded by a weight W = 27 kN through the arrangement shown in the figure. The cable passes over a small frictionless pulley at *B* and is attached at *E* to the end of the vertical arm.

Calculate the axial force *N*, shear force *V*, and bending moment M at section C, which is just to the left of the vertical arm. (*Note:* Disregard the widths of the beam and vertical arm and use centerline dimensions when making calculations.)

A simply supported beam AB supports a trapezoidally distributed load (see figure). The intensity of the load varies linearly from 50 kN/m at support A to 30 kN/m at support B.

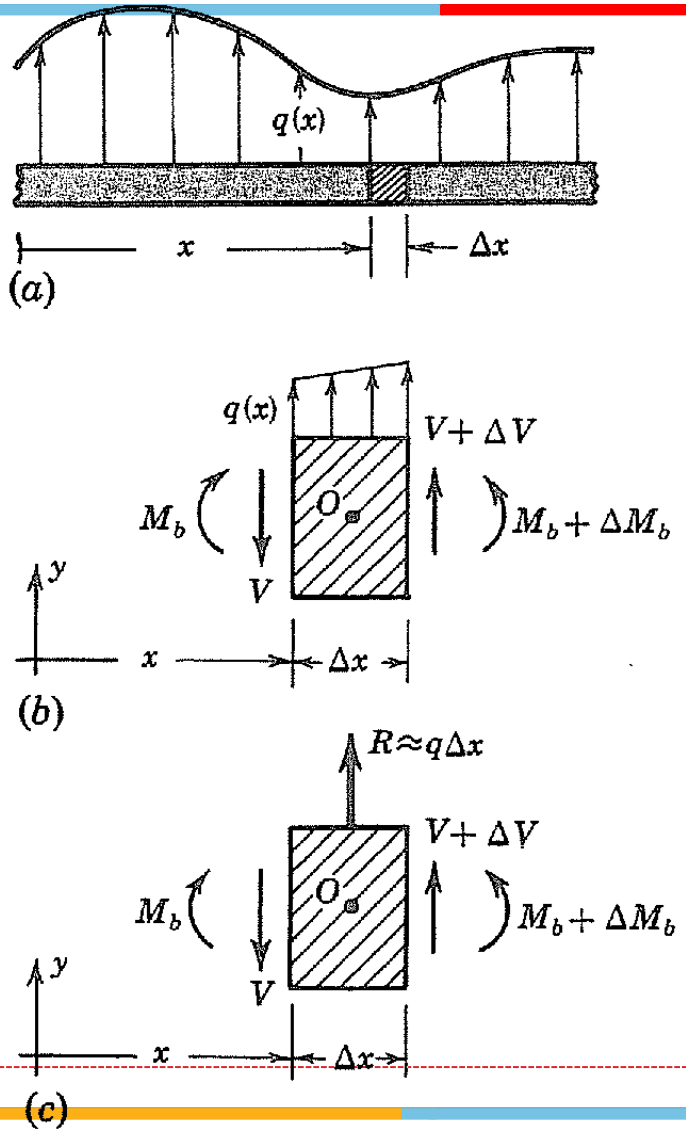
Calculate the shear force V and bending moment M at the midpoint of the beam.

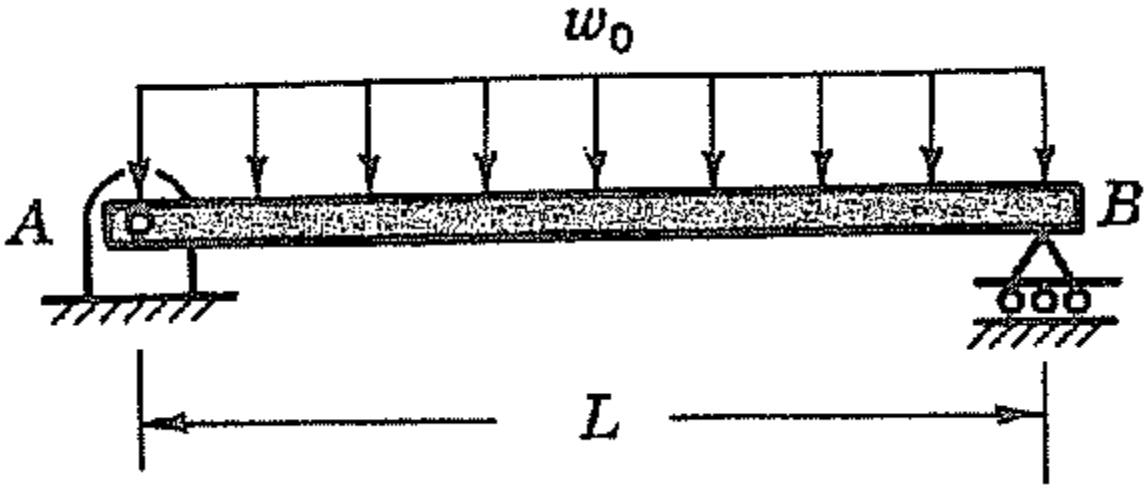




DRAW SFD AND BMD

Differential relationship between Load Intensity, Shear Force and Bending Moment




w_0
A
B
L

Singularity Functions

When calculating SF and BM s for more complex loadings across discontinuities such as concentrated loads and moments, simple methods are not enough.

For handling these more complicated cases, singularity functions provide a more convenient method.

$$f_n(x) = \langle x - a \rangle^n \text{ or } f_n(x) = \langle x - a \rangle_n$$

–Where

- n = any integer (+ve or –ve) including 0
- a = distance on x-axis along the beam, from the selected origin, identifying the location of discontinuity.



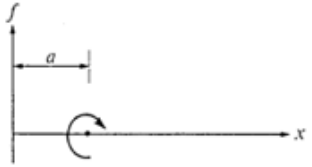
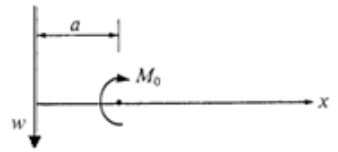
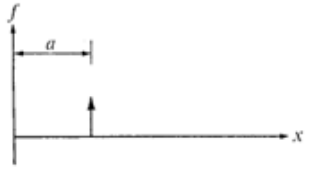
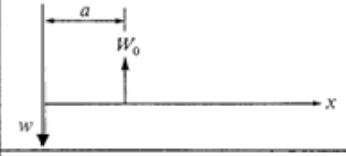
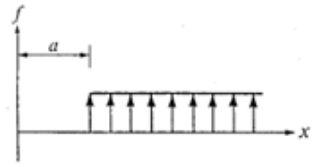
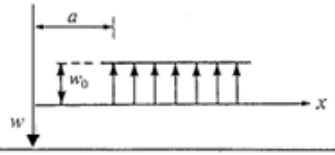
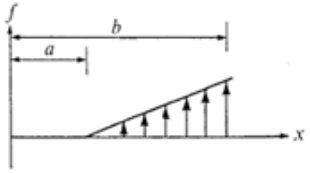
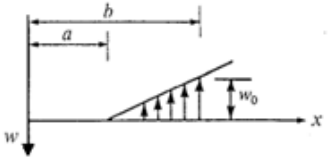
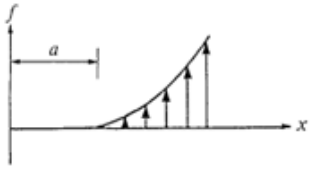
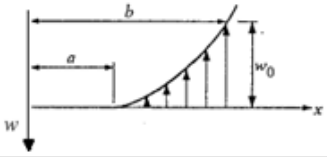
Integration rules for Singularity Functions

$$\text{If } n \geq 0, \int_{-\infty}^{x} \langle x-a \rangle^{n} \, dx = \frac{\langle x-a \rangle^{n+1}}{n+1}$$

$$\text{If } n < 0, \int_{-\infty}^{x} \langle x-a \rangle^{n} \, dx = \langle x-a \rangle^{n+1}$$

Family of Singularity Functions

innovate achieve lead

Singularity function	Graphical representation	Loading	
$f_{-2}(x) = \langle x-a\rangle^{-2}$ (couple)		$w(x) = M_0\langle x-a\rangle^{-2}$	
$f_{-1}(x) = \langle x-a\rangle^{-1}$ (concentrated load)		$w(x) = W_0\langle x-a\rangle^{-1}$	
$f_0(x) = \langle x-a\rangle^{0}$ (uniformly distributed load)		$w(x) = w_0\langle x-a\rangle^{0}$	
$f_1(x) = \langle x-a\rangle^{1}$ (linearly distributed load)		$w(x) = \frac{w_0}{b-a}\langle x-a\rangle^{1}$	
$f_2(x) = \langle x-a\rangle^{2}$ (quadratic distributed load)		$w(x) = \frac{w_0}{(b-a)^2}\langle x-a\rangle^{2}$	

Rules for Singularity Functions

Rule #1:

$$\langle x-a \rangle^0 = \begin{cases} 1 & \text{when } x \geq a \\ 0 & \text{when } x < a \end{cases}$$

$$\langle x-a \rangle^n = \begin{cases} (x-a)^n & \text{when } x \geq a \\ 0 & \text{when } x < a \end{cases}$$

Rule #2:

1. Distributed load w(x) is zero order: e.g. $w_o\langle x-a\rangle^o$
2. Pointed load P(x) is (-1) order: e.g. $P\langle x-a\rangle_{-1}$
3. Moment M is (-2) order: e.g. $M_o\langle x-a\rangle_{-2}$

Rule #3:

$$\int \langle x-a \rangle^{-2} dx = \langle x-a \rangle^{-1}$$

$$\int \langle x-a \rangle^{1} dx = \langle x-a \rangle^{2}$$

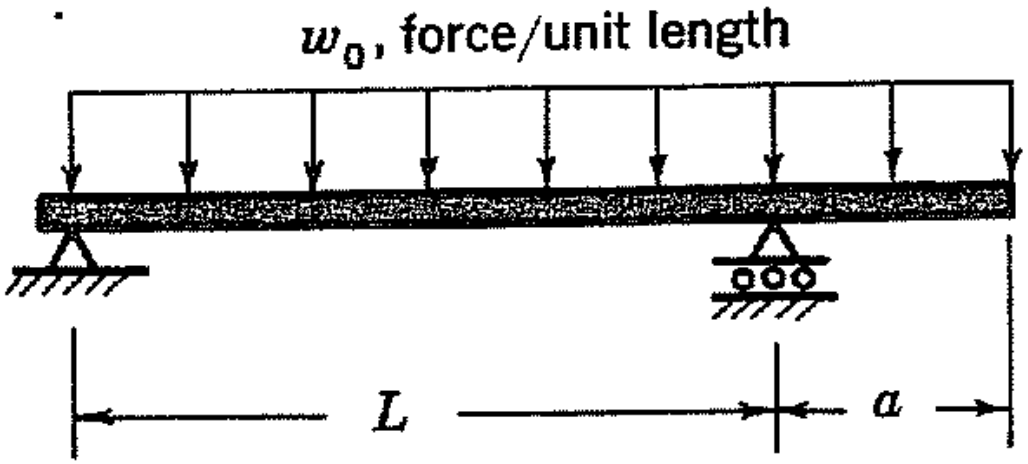
$$\int \langle x-a \rangle^{0} dx = \langle x-a \rangle^{1}$$

$$\int \langle x-a \rangle^{1} dx = \frac{1}{2} \langle x-a \rangle^{2}$$

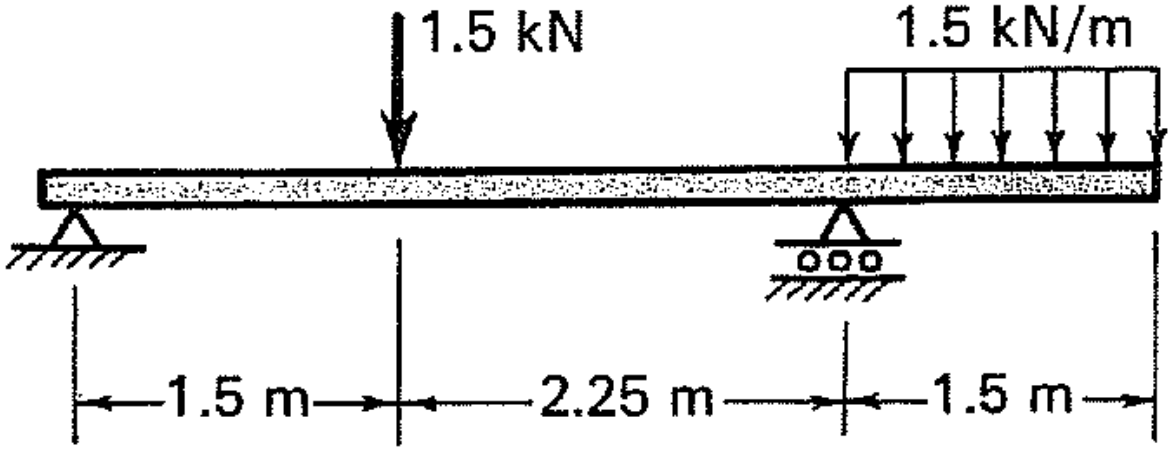
$$\int \langle x-a \rangle^{2} dx = \frac{1}{3} \langle x-a \rangle^{3}$$

Rule #4:

1. Set up w = w(x) first, by including all forces, from the left to the right of the beam.
2. Integrating w once to obtain V, w/o adding any constants.
3. Integrating V to obtain M, w/o adding any constants.

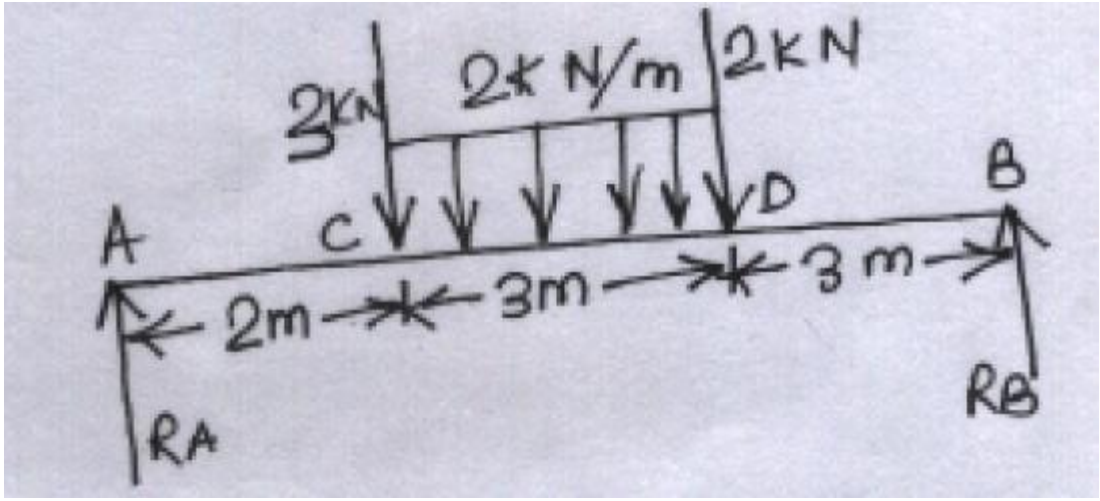


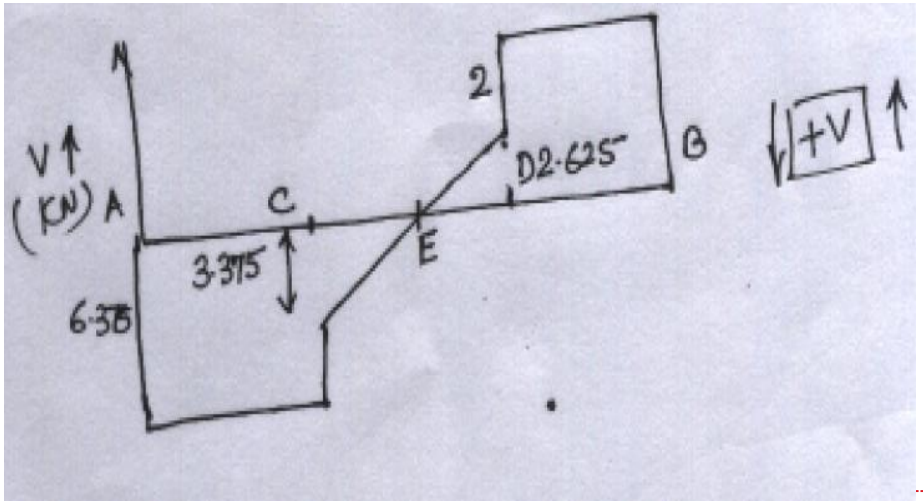
Prob. 3.7

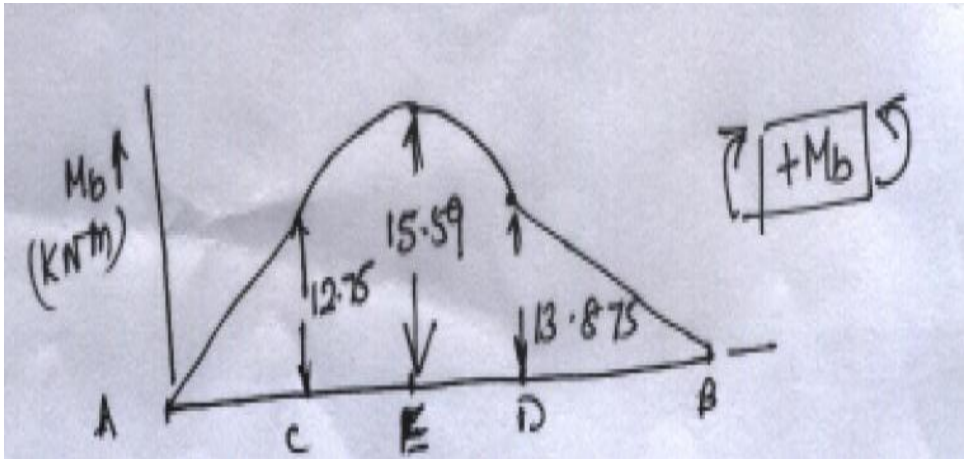


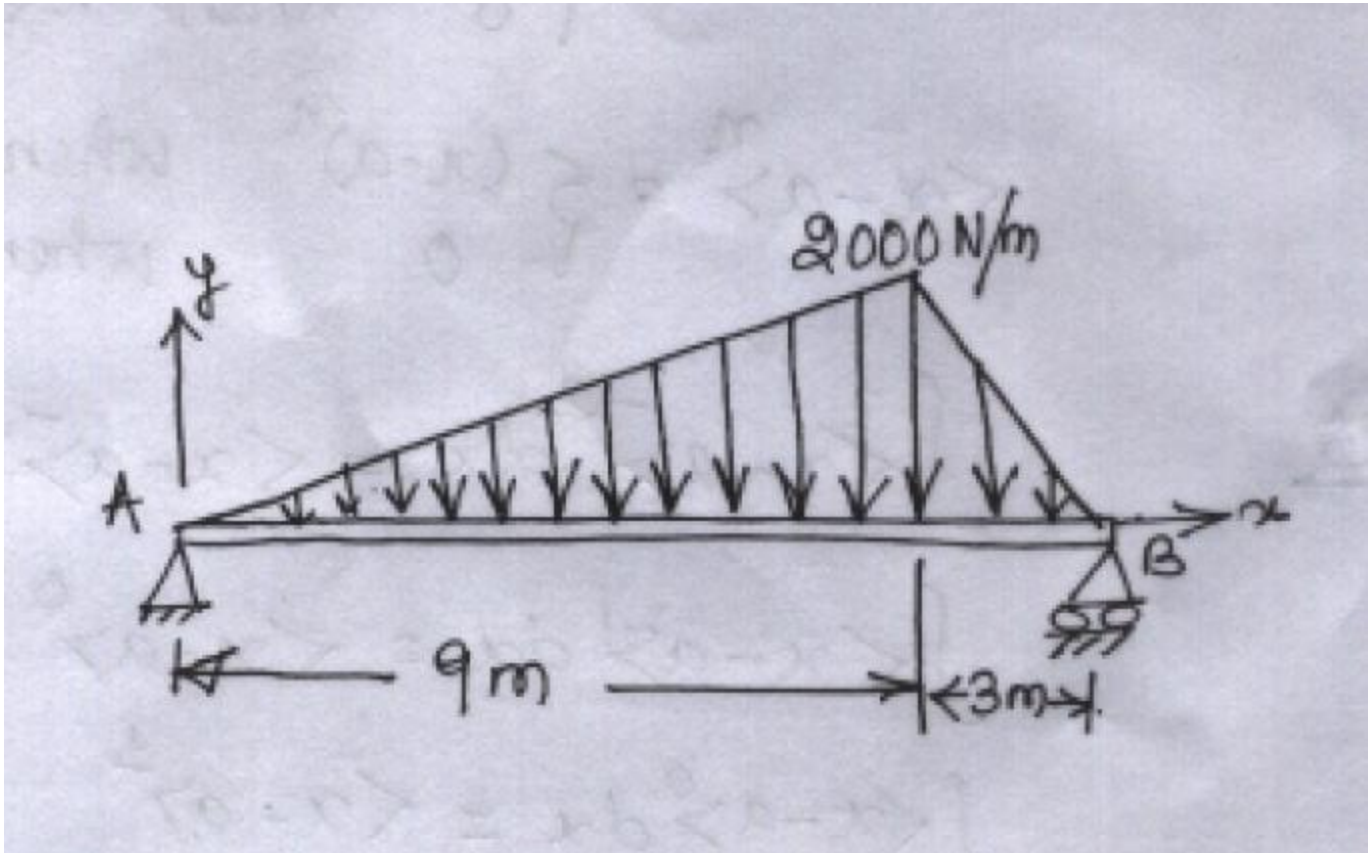
Prob. 3.8

Draw SFD & BMD. Determine the position and magnitude of maximum B.M.








2000 N/m
y
A
B
x
9m
3m


THANK YOU FOR YOUR ATTENTION